Highway Engineering

- INS3441 -

PROBLEM SESSION 7



Question – 1

On a road section with 2x2 lane type and design speed of 90km/h, two vertical alignments with slopes g_1 = 0% and g_2 = 5.58% intersect at a point P. It is planned that a parabolic vertical curve is design to be placed between the vertical alignments. The values of intersection point are P_{km} = (4+350) and $P_{elevation}$= 435 m. Max speed of the heavy vehicles are given as 60 km/h due to slope restrictions.

a) Calculate the elevation and km values of the beginning, ending, mid and inflection points of the vertical curve.

b) What would happen to the length of the vertical curve if the road was designed as undivided road type.

Given:

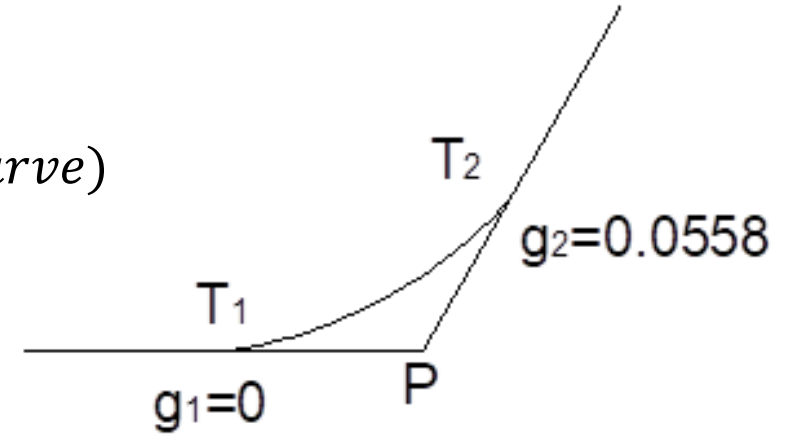
$$d_i = 0.3 * V_i + 8 \qquad f = 0.35$$

$$S > L \rightarrow L = 2 * S - \left(\frac{1.22 + 0.035 * S}{G}\right) \qquad t_r = 1.5\ s$$

$$S < L \rightarrow L = \frac{G * S^2}{1.22 + 0.035 * S}$$

a) Calculate the elevation and km values of the beginning, ending, mid and inflection points of the vertical curve.

$g_1 = \%0$
$g_2 = \%5.58$ $\rightarrow$ $G = g_1 - g_2 = 0 - 0.0558 = -0.0558$ $(sag\ vertical\ curve)$



If it is divided road; ***S = max(l_s ; l_{fe})***

$$l_s = \frac{d_1 + d_2}{V_1 - V_2} * V_1 = \frac{35 + 26}{90 - 60} * 90 = 183\ m$$

$P_{km} = (4 + 350)$
$P_{\text{elev}} = 435\ m$

$d_i = 0.3 * V_i + 8$

$d_1 = 0.3 * 90 + 8 = 35$ m

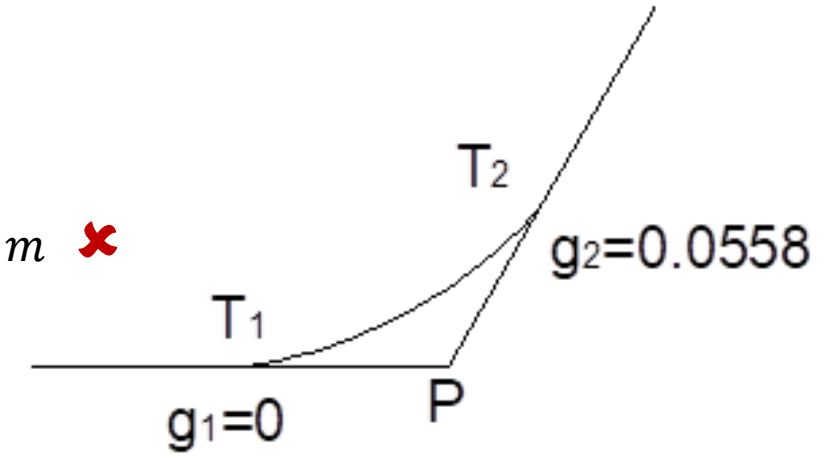
$d_2 = 0.3 * 60 + 8 = 26$ m

$$S = max(l_s; l_{fe}) = 183\ m$$

$$l_{fe} = 0.278 * V * t_r + 0.00394 * \frac{V^2}{f \mp s} = 0.278 * 90 * 1.5 + 0.00394 * \frac{(90)^2}{0.35 - 0.0558} = 146\ m$$

$$S = 183\ m$$

$$S > L \rightarrow L = 2 * S - \frac{1.22 + 0.035 * S}{G} = 2 * 183 - \frac{1.22 + 0.035 * 183}{0.0558} = 229.35\ m$$ ✘



$$S < L \rightarrow L = \frac{G * S^2}{1.22 + 0.035 * S} = \frac{0.0558 * (183)^2}{1.22 + 0.035 * 183} = 245.07\ m$$ ✔

$$L = 245.07 \cong 245\ m \rightarrow t = \frac{L}{2} = \frac{245}{2} = 122.5\ m \qquad e = \frac{G * L}{8} = \frac{0.0558 * 245}{8} = 1.71\ m$$

Km Calculation	Elevation Calculation
$T_{1km} = (4 + 350) - 122.5 = 4 + 227^{50}$	$T_{1elev} = P_{elev} = 435\ m$
$T_{2km} = (4 + 350) + 122.5 = 4 + 472^{50}$	$T_{2elev} = 435 + 122.5 * 0.0558 = 441.84\ m$
$B_{km} = P_{km} = 4 + 350$	$B_{elev} = 435 + 1.71 = 436.71\ m$
$DN_{km} = T_{1km} = 4 + 227^{50}$ OR	$DN_{elev} = T_{1elev} = 435\ m$ OR
$DN_{km} = (4 + 227^{50}) + \frac{0.0 * 245}{0.0558} = 4 + 227^{50}$	$DN_{elev} = 435 + \frac{(0.0)^2 * 245}{2 * 0.0558} = 435\ m$

b) What would happen to the length of the vertical curve if the road was designed as undivided road type.

If it is undivided road; $\boldsymbol{S = L_s}$

$$Ls = \frac{d_1 + d_2}{V_1 - V_2} * (V_1 + V_3)$$

$$L_s = \frac{35 + 26}{90 - 60} * (90 + 90) = 366\ m$$

$$\boxed{S = Ls = 366\,m}$$

$$S > L \rightarrow L = 2 * 366 - \frac{1.22 + 0.035 * 366}{0.0558} = 480.57\ m \rightarrow S = 366 \not> L = 480.57$$

$$S < L \rightarrow L = \frac{0.0558 * 366^2}{1.22 + 0.035 * 366} = 532.77\ m \rightarrow S = 366 < L = 532.77$$

L is changed as 532.77 m

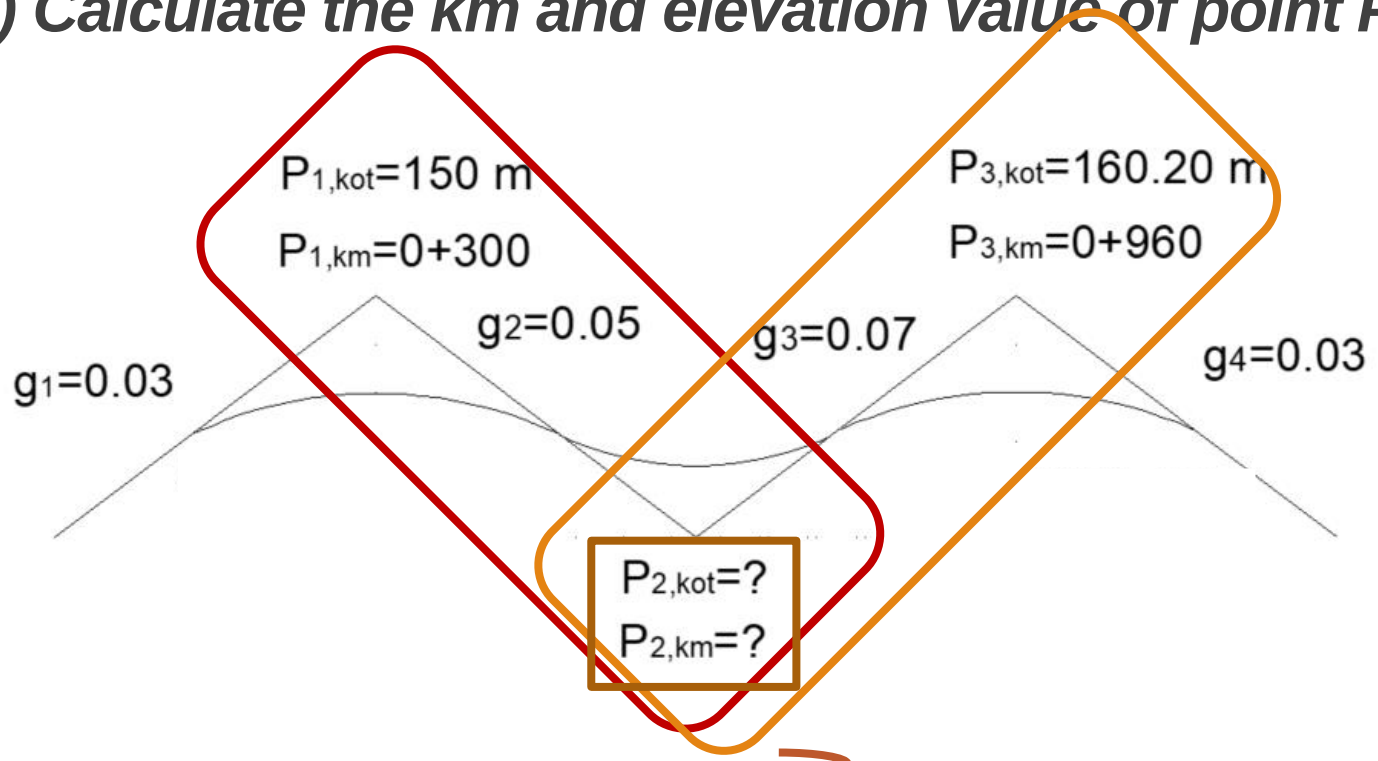
Question – 2

On a profile, the slopes of consecutive vertical alignments are given as $g_1 = 0.03$, $g_2 = -0.05$, $g_3 = 0.07$ and $g_4 = -0.03$. First and second alignments intersect at point P1 and Third and fourth alignments intersect at point P3. The values of intersection points are $P_{1km} = (0+300)$, $P_{1elev} = 150$ m, $P_{3km}=(0+960)$, $P_{3elev}=160.20$ m. Also the following vertical curves exactly start at the end point of the previous vertical curves.

a) Calculate the km and elevation value of point P_2.

b) Calculate the radius and lengths of the curves. It is required that all of the vertical curves will be circular and the radius of first and second curves will be equal.

a) Calculate the km and elevation value of point P_2.



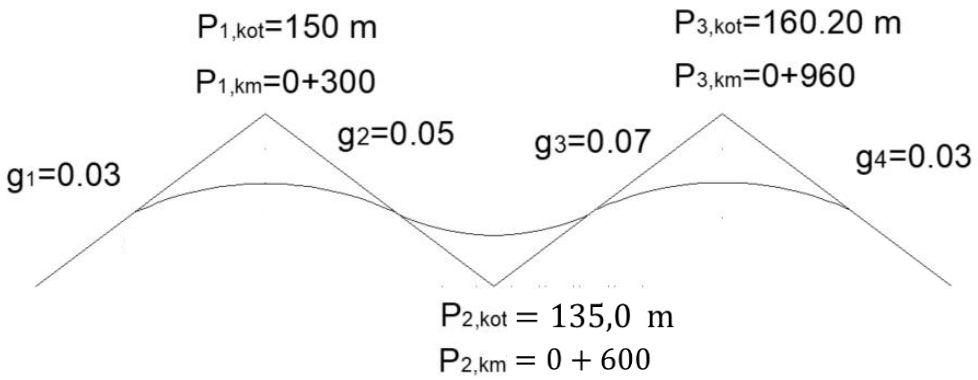
$$-0.05 = \frac{P_{2elev} - 150}{P_{2km} - 300}$$

$$0.07 = \frac{160.2 - P_{2elev}}{960 - P_{2km}}$$

$$P_{2km} = 0 + 600$$

$$P_{2elev} = 135.0 \text{ m}$$

b) Calculate the radius and lengths of the curves. It is required that all of the vertical curves will be circular and the radius of first and second curves will be equal.



$$G_1 = 0.03 - (-0.05) = 0.08$$

$$G_2 = -0.05 - 0.07 = -0.12 \quad \rightarrow 0.12$$

$$G_3 = 0.07 - (-0.03) = 0.10$$

$$\mathbf{R_1 = R_2 = R}$$

$$\frac{L_1}{2} + \frac{L_2}{2} = P_{2km} - P_{1km} = 300 \ \text{m}$$

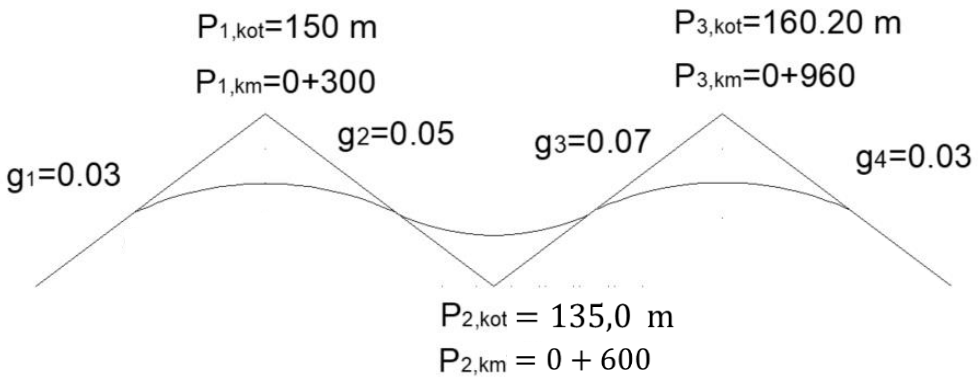
$$L_1 + L_2 = 600 \ \text{m}$$

$$L_1 = G_1 * R_1 = 0.08 * R_1$$

$$L_2 = G_2 * R_2 = 0.12 * R_2$$

$$(0.08 + 0.12) * R = 600 \ \text{m} \quad \rightarrow \ R = 3000 \ \text{m} = R_1 = R_2$$

$$L_1 = 0.08 * 3000 = 240 \ \text{m} \qquad L_2 = 0.12 * 3000 = 360 \ \text{m}$$



$$G_1 = 0.03 - (-0.05) = 0.08$$

$$G_2 = -0.05 - 0.07 = -0.12 \quad \rightarrow 0.12$$

$$G_3 = 0.07 - (-0.03) = 0.10$$

$$\frac{L_2}{2} + \frac{L_3}{2} = P_{3km} - P_{2km}$$

$$L_2 + L_3 = 2 * (960 - 600) = 720 \ \text{m}$$

$$360 + L_3 = 720 \ \text{m} \longrightarrow L_3 = 360 \ \text{m}$$

$$R_3 = \frac{L_3}{G_3} = \frac{360}{0.10} = 3600 \ \text{m}$$

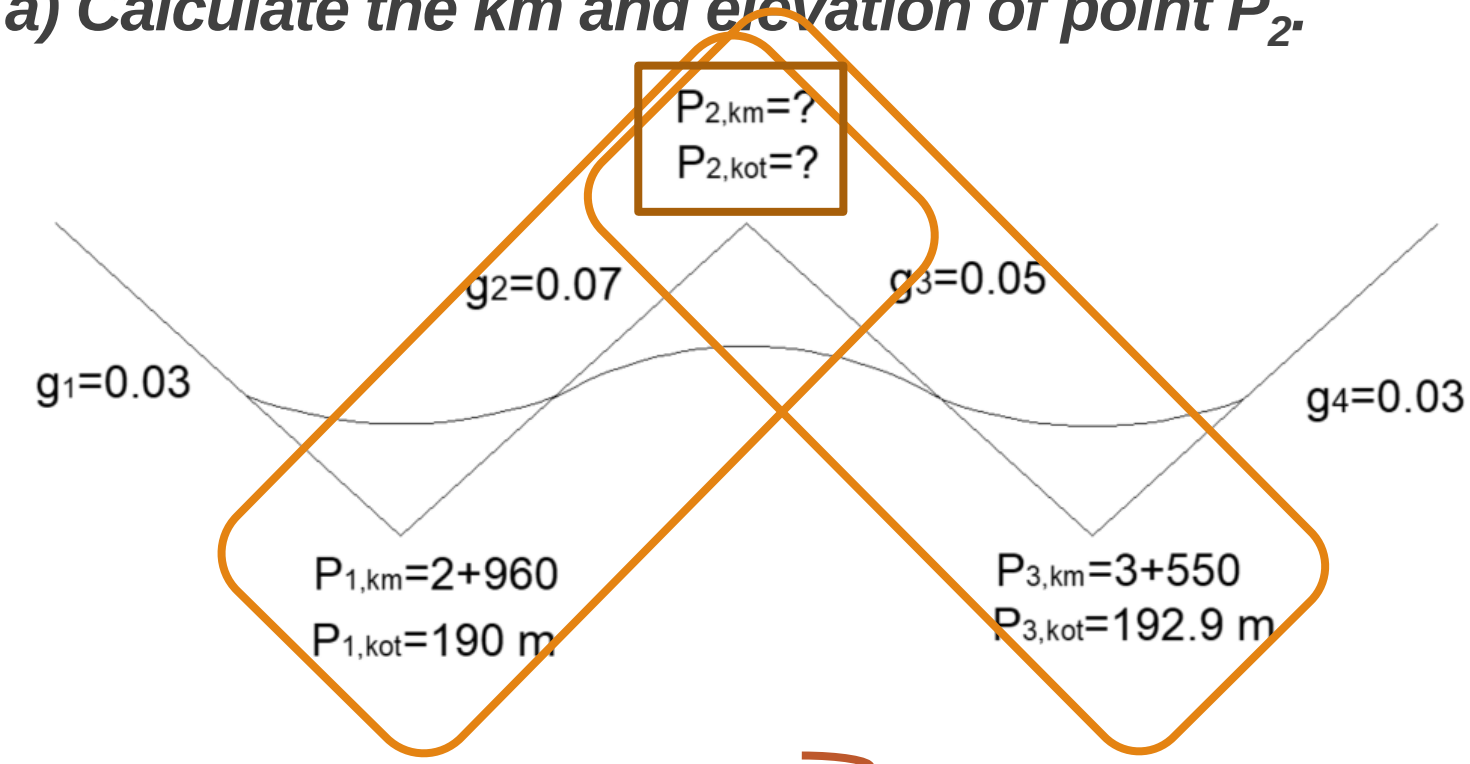
Question – 3

On a profile, the slopes of consecutive vertical alignments are given as $g_1 = -0.03$, $g_2=0.07$, $g_3=-0.05$ and $g_4= 0.03$. First and second alignments intersect at point P1. Third and fourth alignments intersect at point P3. The values of intersection points are $P_{1km} = (2+960)$, $P_{1elev} = 190$ m, $P_{3km}=(3+550)$, $P_{3elev}=192.90$ m. Also the following vertical curves exactly start at the end point of the previous vertical curves.

a) Calculate the km and elevation of point P_2.

b) Calculate the radius and lengths of the curves. It is required that the vertical curves will be circular and the radius of first and second curves will be related as $\mathbf{R_1 = 1.5 * R_2}$.

a) Calculate the km and elevation of point P_2.



$$0.07 = \frac{P_{2elev} - 190}{P_{2km} - 2960}$$

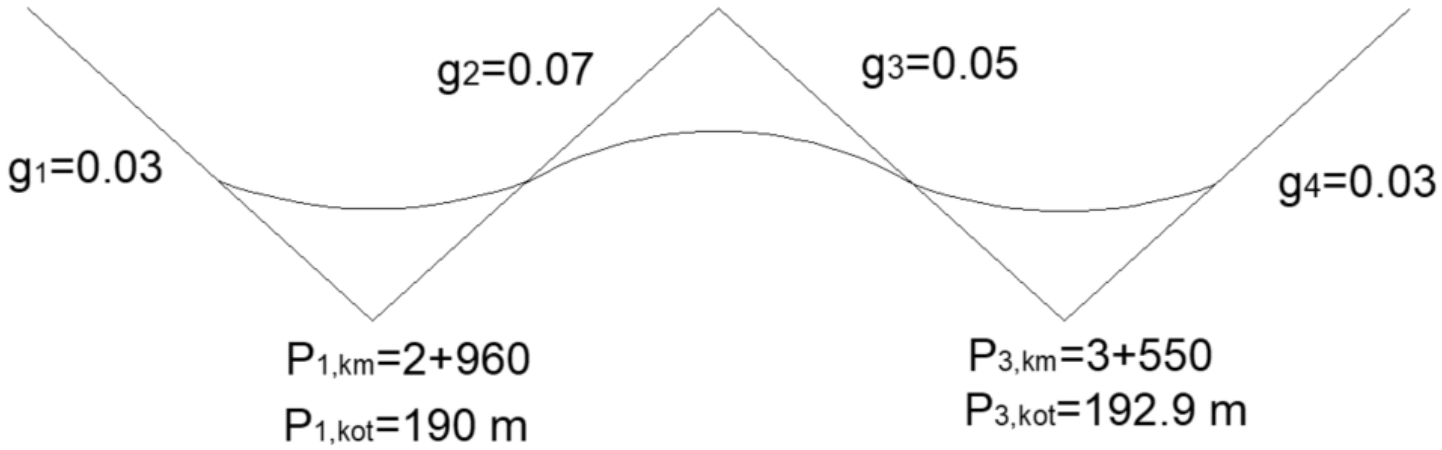
$$0.05 = \frac{P_{2elev} - 192.9}{P_{2km} - 3550}$$

$$P_{2km} = 3 + 230$$

$$P_{2elev} = 208.9 \text{ m}$$

b) Calculate the radius and lengths of the curves. It is required that the vertical curves will be circular and the radius of first and second curves will be related as $R_1 = 1.5 * R_2$

$$P_{2km} = 3 + 230$$
$$P_{2kot} = 208{,}9 \ \text{m}$$



$$G_1 = 0{,}10$$
$$G_2 = 0{,}12$$
$$G_3 = 0{,}08$$

$$\frac{L_1}{2} + \frac{L_2}{2} = (3 + 230) - (2 + 960) = 270$$

$$L_1 + L_2 = 540 \ \text{m}$$

$$L_1 = G_1 * R_1 = 0.10 * (1.5 * R_2) = 0.15 * R_2$$

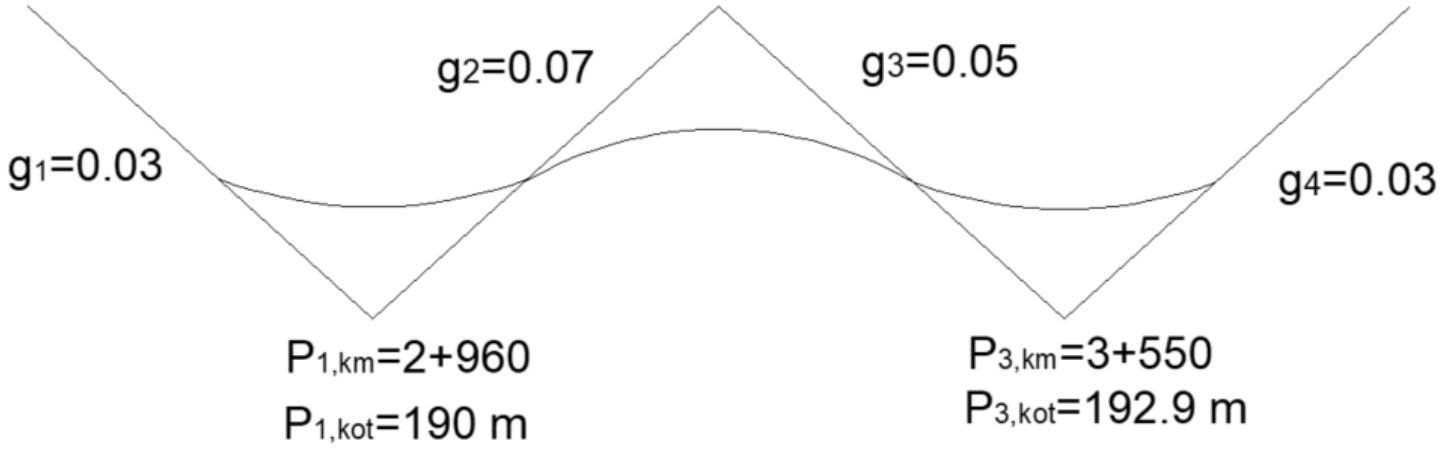
$$L_2 = G_2 * R_2 = 0.12 * R_2$$

$$(0.15 + 0.12) * R_2 = 540 \ \text{m}$$

$$R_2 = 2000 \ \text{m}$$

$$R_1 = 3000 \ \text{m}$$

$$P_{2km} = 3 + 230$$
$$P_{2kot} = 208{,}9 \text{ m}$$



$$G_1 = 0{,}10$$
$$G_2 = 0{,}12$$
$$G_3 = 0{,}08$$

$$L_1 = 0.10 * 3000 = 300 \text{ m}$$

$$L_2 = 0.12 * 2000 = 240 \text{ m}$$

$$L_2 + L_3 = 640 \text{ m}$$

$$L_3 = 640 - 240 = 400 \text{ m}$$

$$\frac{L_2}{2} + \frac{L_3}{2} = P_{3km} - P_{2km} = (3 + 550) - (3 + 230) = 320 \text{ m}$$

$$R_3 = \frac{L_3}{G_3} = \frac{400}{0.08} = 5000 \text{ m}$$

Question – 4

A state highway with two directions and double lane is designed for design speed of 90 km/h. The data about a section of this highway and some other parameters are provided below. Half of the centrifugal force will be compensated by superelevation.

Point No	Km	Grade Elevation	Project Criteria
120	1+550	610 m	Platform Width = 10 m, Max Superelevation = %8
Φ_7	1+700	-	Crown Slope=%2, Speed of overpassed veh.=50km/h
P (point of intersection)	2+250	624 m	μ_r= 0.03, f=0.40, d_i= 0.2*V_i+8 , Q_m=2/3*Q_t ,t_r=1 s
132	3+000	594 m	q_{theo}= 0.786*(V_p^2 /R), p' = V^3/(46.7*R*L) < 0.6 m/s^3
			S < L → L= ($G*S^2$)/11, S > L → L= 2*S - (11/G)

Question – 4

a) Find the slopes between points (120)-P and P-(132).

b) According to the zero line study and route study two alignments intersect at an S point with $\Delta=(57.30)^o$ curve angle. Calculate all other components of an horizontal curve with Radius 250 m. (D=?, t=?, b=?).

c) Select the clothoid with largest possible parameter or this curve, calculate its length and draw constant centerline superelevation with ramp diagram showing all values by making necessary calculations.

d) Calculate the elevations of the inner and outer edges of point (120).

e) Calculate the length of a parabolic vertical curve which can be placed between the grade elevation alignments with slope values g_1 and g_2 which intersects at point P. Find km values and elevations of tangents, mid and inflection points.

f) Check if this highway was designed as 2x2 lanes would it change the vertical curve length.

g) Calculate the limits of the traction force which is needed for a vehicle to re-start its motion (Qt =12 tons) after stopping.

a) Find the slopes between points (120)-P and P-(132).

$$s = \frac{\Delta h}{\Delta L}$$

$$s_1 = \frac{624 - 610}{(2+250) - (1+550)} = \%2 \qquad s_2 = \frac{594 - 624}{(3+000) - (2+250)} = -\%4$$

b) According to the zero line study and route study two alignments intersect at an S point with Δ =(57.30)º curve angle. Calculate all other components of an horizontal curve with Radius 250 m. (D=?, t=?, b=?).

$$D = \frac{2\pi R\Delta}{360} = \frac{2 * \pi * 250 * 57,30}{360} = 250m$$

$$t = R * tg(\frac{\Delta}{2}) = 250 * tg(\frac{57,30}{2}) = 136,59m \qquad b = R * (\sec\frac{\Delta}{2} - 1) = 34,88m$$

c) Select the clothoid with largest possible parameter or this curve, calculate its length and draw constant centerline superelevation with ramp diagram showing all values by making necessary calculations.

Superelevation

$$q_{pra} = 0,5 * q_{teo} = 0,5 * 0,786 * \frac{V_p^2}{R} = 0,5 * 0,786 * \frac{90^2}{250} = 12,73 > 8$$

Speed limit

$$8 = 0,5 * 0,786 * \frac{V_p^2}{250} \Rightarrow V_p = 71,34 km / sa$$

Optical condition:

$$\frac{R}{3} \leq A \leq R \Rightarrow \frac{250}{3} \leq A \leq 250$$

Dynamic condition:

$$A_{\min} = 0,19 * \sqrt{V_p^3} = 0,19 * \sqrt{71,34^3} = 114,48$$

It is asked for maximum possible A parameter so A =250

$$L_g = \frac{A^2}{R} = \frac{250^2}{250} = 250m$$

Minimum length control:

$L_g = 250\ m > 45\ m$ satisfied

Geometrical control:

$$\frac{L_g}{2} \le \frac{D}{2} \rightarrow \frac{250}{2} \le \frac{250}{2}$$ satisfied

Jerk Condition:

$$p' = \frac{V_p^3}{46,7 * R * L_g} = \frac{71,34^3}{46,7 * 250 * 250} = 0,12m/s^3 < 0,6m/s^3$$ satisfied

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1}$$

Superelevation at K

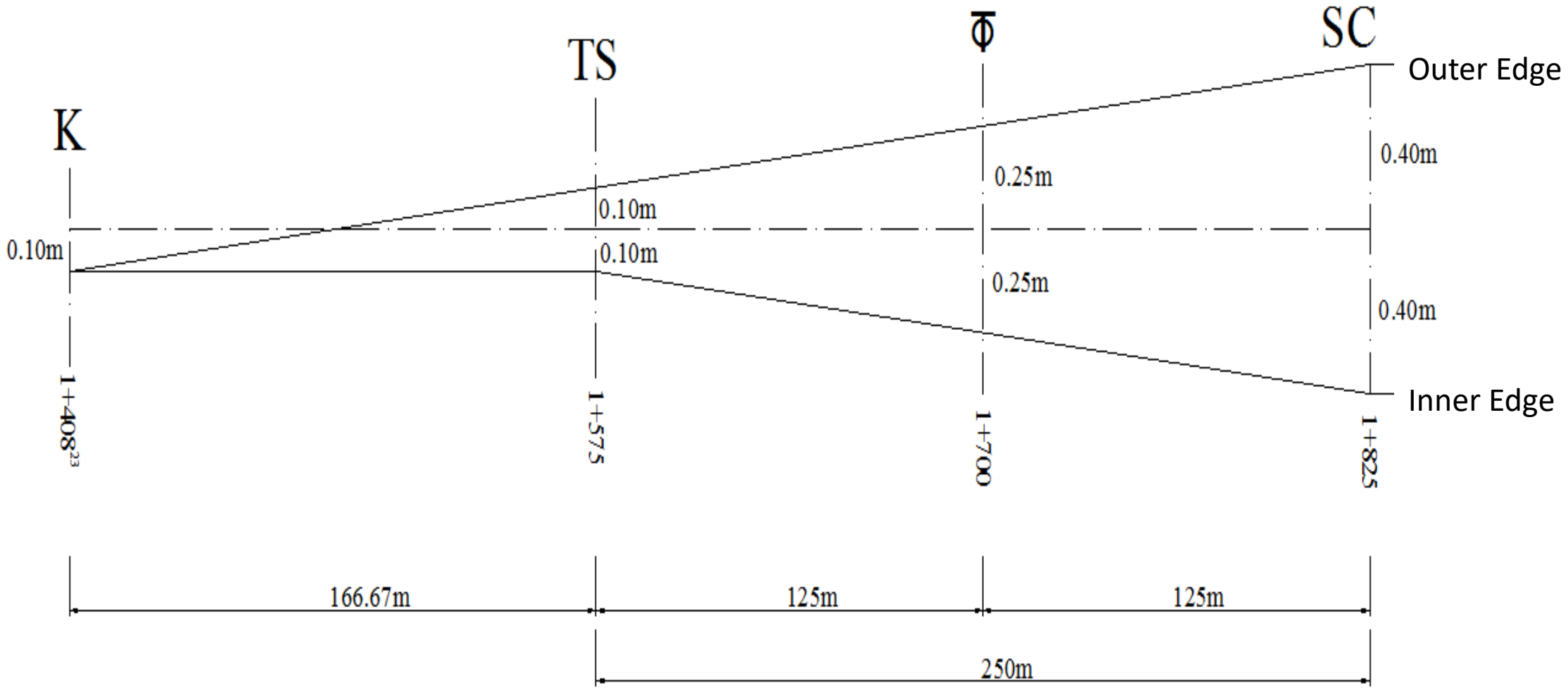
$$h_1 = q * \frac{b}{2} = 0,02 * \frac{10}{2} = 0,10m$$

Superelevation at SC *:*

$$h_2 = q * \frac{b}{2} = 0,08 * \frac{10}{2} = 0,4m$$

Ramp length:

$$K = \frac{2 * h_1 * L_g}{h_2 - h_1} = \frac{2 * 0,1 * 250}{0,4 - 0,1} = 166,67m$$



$$q_{\bar{\Phi}} \rightarrow \frac{x+2}{8+2} = \frac{291.67}{416.67} \quad \Rightarrow q_{\bar{\Phi}} = \%5$$

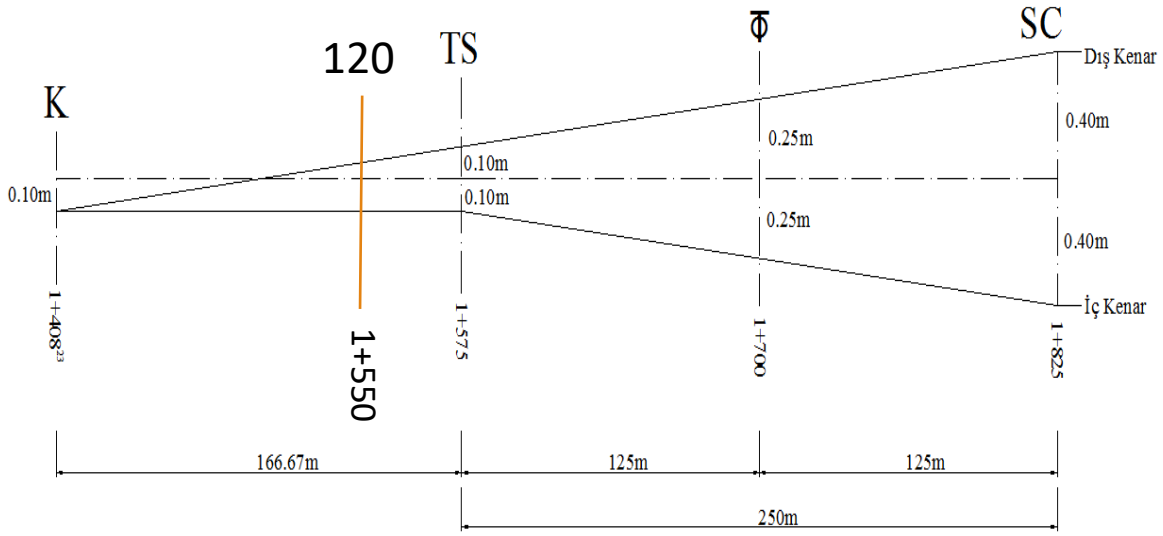
$$h_{\bar{\Phi}} = 0.05 * \frac{10}{2} = 0.25\ m$$

d) Calculate the elevations of the inner and outer edges of point (120).

$$\frac{0,4-(-0,1)}{166,67+250}=\frac{h_{120}-(-0,1)}{(1+550)-(1+408^{33})}=0,07m$$

Outer edge = 610 + 0,07 = 610,07 m

Inner edge = 610 – 0,1 = 609,90 m



e) Calculate the length of a parabolic vertical curve which can be placed between the grade elevation alignments with slope values g_1 and g_2 which intersects at point P. Find km values and elevations of tangents, mid and inflection points.

$d_1 = 8 + 0{,}2*90 = 26$ m

$d_2 = 8 + 0{,}2*50 = 18$ m

S=Ls ➔ $S = \frac{d_1 + d_2}{V_1 - V_2} * (V_1 + V_3) = \frac{26+18}{90-50} * (90+90) = 198m$

$G = 0{,}02 - (-0{,}04) = 0{,}06$

S > L: $L = 2S - \frac{11}{G} = 2*198 - \frac{11}{0{,}06} = 212{,}67m$

S < L: $L = \frac{GS^2}{11} = \frac{0{,}06*198^2}{11} = 213{,}81m$ ✓

Let's use as 215 m for this question

$$t = \frac{L}{2} = \frac{215}{2} = 107,5m$$

$$e = \frac{GL}{8} = \frac{0,06*215}{8} = 1,61m$$

Km Calculations
B(km) = P(km) = 2 + 250
T_1(km) = B(km) – t = (2 + 250) – 107,5 = $2+142^{50}$
T_2(km) = B(km) + t = (2 + 250) + 107,5 = $2+357^{50}$
$DN(km) = T_1(km) + \frac{g_1 * L}{G} = 2 + 142^{50} + \frac{0,02*215}{0,06} = 2 + 214^{16}$

Elevation Calculations
B(km) = P(elev) – b = 624 – 1,61 = 622,39 m
T_1(elev) = P(elev) – t*g_1 = 624 – 107,5*0,02 = 621,85 m
T_2(elev) = P(elev) + t*g_2 = 624 + 107,5*(-0,04) = 619,70 m
$DN(kot) = T_1(kot) + \frac{g_1^2 * L}{2G} = 621,85 + \frac{0,02^2 * 215}{2*0,06} = 622,56m$

f) Check if this highway was designed as 2x2 lanes would it change the vertical curve length.

S = max(l_{fe}, l_s)

$$L_{fe} = 0,278 * V * t_r + \frac{0,00394 * V^2}{f \pm s} = 0,278 * 90 * 1 + \frac{0,00394 * 90^2}{0,4 - 0,04} = 114m$$

$$L_s = \frac{26 + 18}{90 - 50} * (90) = 99m$$

S=114 m

$$S > L : \; L = 2S - \frac{11}{G} = 2 * 114 - \frac{11}{0,06} \cong 45m$$ ✓

L_{min} = 120 m for vertical curves; so L = 120 m

g) Calculate the limits of the traction force which is needed for a vehicle to re-start its motion (Qt =12 tons) after stopping.

$$\sum W = W_y + W_k + W_h + W_e + W_i$$

V = 0; so $W_0=0$, $W_k=0$ and $W_i=0$

$$\sum W = W_y + W_e$$

$$W_y = \mu_r * Q = 0,03*12000*10 = 3600N$$

$$W_e = Q * s = 12000*10*0,04 = 4800N$$

$$\sum W = W_y + W_e = 3600 + 4800 = 8400N$$

$$Q_m * f = 12000 * \frac{2}{3} * 10 * 0,4 = 32000N$$

$$8400 \le Z_{TR} \le 32000$$

$$Q_m * f \ge Z_{tr} \ge \Sigma W$$